# A Novel Method for the Automatic Grading of Retinal Vessel Tortuosity

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***Abstract*—Tortuosity is among the first alterations in the retinal vessel network to appear in many retinopathies, such as those due to hypertension. An automatic evaluation of retinal vessel tortuosity would help the early detection of such retinopathies. Quite a few techniques for tortuosity measurement and classification have been proposed, but they do not always match the clinical concept of tortuosity. This justifies the need for a new definition, able to express in mathematical terms the tortuosity as perceived by ophthalmologists. We propose here a new algorithm for the evaluation of tortuosity in vessels recognized in digital fundus images. It is based on partitioning each vessel in segments of constant-sign curvature and then combining together each evaluation of such segments and their number. The algorithm has been compared with other available tortuosity measures on a set of 30 arteries and one of 30 veins from 60 different images. These vessels had been preliminarily ordered by a retina specialist by increasing perceived tortuosity. The proposed algorithm proved to be the best one in matching the clinically perceived vessel tortuosity.**

***Index Terms*—Curvature, fundus imaging, hypertensive retinopathy, retina, tortuosity, vessels.**

## I. Introduction

THE CLINICAL observation of the ocular fundus is an important tool for identifying and monitoring many pathological manifestations coming from the major cardiovascular, central nervous and endocrine-metabolic system diseases [1]. Most retinopathies, e.g., from hypertension or diabetes, could be diagnosed early and effectively treated if an accurate and objective analysis of symptoms at their initial onset could be performed. The analysis should be sensitive enough to detect minor pathological signs, and robust enough to be able to compare results with accepted clinical standards and with results obtained from the same patient at different times. This latter requirement is of paramount importance when assessing the effect of therapeutic treatments and even more when evaluating in a quantitative way the efficacy of new drugs during their development.

In evaluating hypertensive retinopathy, when no major signs of retinal lesion are present (such as cotton wool spots, haemorrhages, exudates), the clinical diagnostic procedure involves a careful evaluation of the main features of the network of retinal vessels, obtained from fundus camera images or ophthalmoscopic examination. Morphological features of the retinal vascular tree are often the early symptoms indicating the onset of some retinopathies [2], or are prognostic indicators for others, as for the retinopathy of prematurity [3].

One of the first changes in vessels morphology to occur is the increase in vessel tortuosity [4]. Tortuosity can be intuitively defined as a nonsmooth appearance of the vessel course: in Fig. 1 two retinas with different vessel tortuosity are shown.

Even if the severity of disease has been proved to be correlated with an increase in vessel tortuosity in retinopathy of prematurity [5], hypertensive retinopathy [6], and some rarer and hereditary retinopathies (e.g., [7]), no objective and quantitative grading has been ever adopted to assess vessel tortuosity. Also in [6], where a five level grading scale was proposed for the evaluation of tortuosity, grading is based on a subjective and qualitative measure. In order to evaluate the clinical significance of tortuosity changes with time, or to compare different levels of the same retinopathy, there is a strong need to develop a tortuosity measure that matches the clinical perception of ophthalmologists.

At first, we have to clearly understand the main factors that influence the classification of a vessel as tortuous or nontortuous. This is even more important in analyzing retinal images, in which, because of the hemispherical shape of the retina, not only straight vessels but also long vessels presenting a smooth semi-circular or parabolic shape are considered as non-tortuous. The need to develop a numerical index incorporating these factors arose from the observation that previously proposed methods (see [8] for a review) fail in differentiating the tortuosity of structures that visually appear to be very different in tortuosity, as it will be shown in Section III-A. Recently, a new measure has been proposed [9], which tries to overcome the limitations of these previous approaches, and we will show how it compares to the one we propose.

Finally, since we aim at the realization of a fully automated system to grade tortuosity of retinal vessels, we eventually would like to use it in combination with one of the many vessel segmentation algorithms proposed (e.g., [10]–[19]). With this objective, we also would like the proposed measure to be robust with respect to possible errors in vessel center-line detection and to the breaking of vessel connectivity.

## II. Tortuosity Properties

Vessel tortuosity measures do not have a formal clinical definition, but common practice has outlined an evaluation of the tortuosity that has to conform to some properties. In order to

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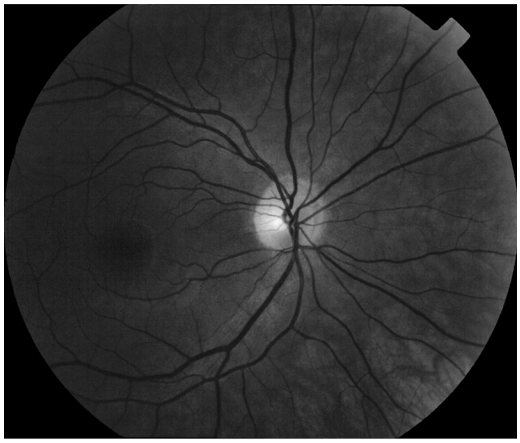
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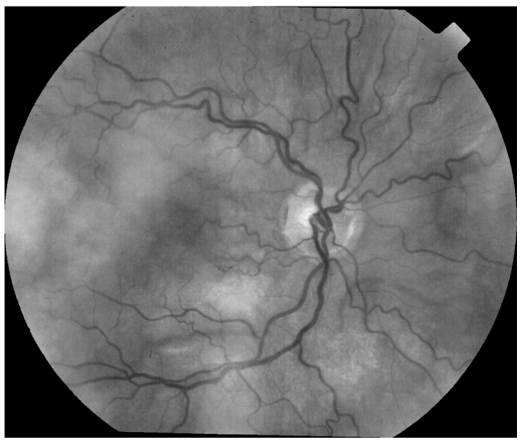
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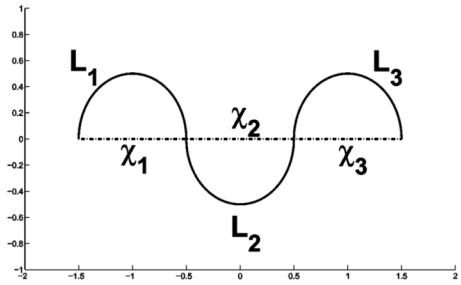


(a)

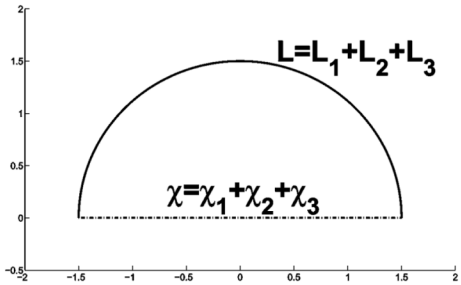


(b)

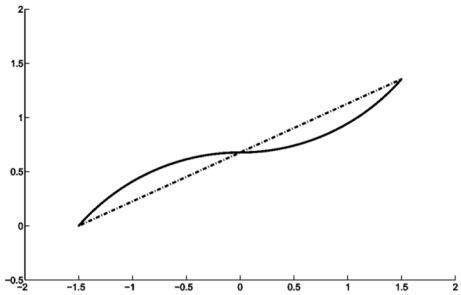
Fig. 1. Two retina images with different vessel tortuosity: image (a) presents a normal/very mild tortuosity, whereas image (b) shows a severe vessel tortuosity (hypertensive patient).




(a)




(b)



(c)

Fig. 2. First two curves [(a) and (b)] have very different tortuosity but the same arc length $L$ and chord length $\chi$. Second and last curves [(b) and (c)] have the same mean angular difference $(\mathrm{MAC})$ despite their different perceived tortuosity. Curve in (c) has a curvature integral of $\pi/2$, whereas the one in (b) has curvature integral $\pi$, even if the one (c) is perceived as more tortuous.

obtain a clinically meaningful vessel tortuosity measure, i.e., a measure that is able to match the ophthalmologist's evaluation, it is necessary to make these properties explicit. Then, using a proper mathematical formulation of the tortuosity functional, the proposed measure must be shown to satisfy these properties.

### *A. Rigid Transformations*

A sound approach to extract properties of a not formally defined index is to perform a preliminary study on the invariance properties of such index with respect to the most relevant transformations in the domain space. In particular, we will consider rigid transformations of a vessel: translation, rotation, and scaling.

Translation and rotation transformations are not supposed to influence the perception of tortuosity. In fact, these transformations are related to the topology and orientation of vessels in the retina, and do not alter in any way the clinical perception of tortuosity. For the purpose of evaluating tortuosity of vessels with similar calibers, which is the aim of this paper, tortuosity perception can be considered invariant to scaling. On the contrary, when tortuosity measures of vessels with different calibers have to be compared or combined (e.g., to provide a global tortuosity index for the entire retinal image), it is still not clear how the single vessel caliber weights in the overall clinician's perception.

### *B. Composition*

Composition properties deal with the behavior of tortuosity perception when two vessel segments are merged into a single one, or when various sections of the same vessel, in general with different tortuosity measures, build up to give the total vessel tortuosity.

Given two adjacent continuous curves $s_1$ and $s_2$, we define the combination of the two as

$$s_3 = s_1 \oplus s_2.$$

Since the two composing curves are supposed to belong to the same vessel, we can assume the continuity of $s_3$ without loss of generality.

In [8], an intuitive empirical principle was proposed for the tortuosity $\tau$ of two curves $s_1$ and $s_2$

$$\tau(s_1) \leq \tau(s_2) \Rightarrow \tau(s_1) \leq \tau(s_1 \oplus s_2) \leq \tau(s_2) \qquad (1)$$

which means that, when composing two curves, the resulting tortuosity is between those of the composing curves. The counterexample in Fig. 2(a) clearly shows that this statement can not be accepted in conjunction with the principle of invariance with respect to rotation and scale: curves $L_1$, $L_2$, and $L_3$ are perceived by themselves as non tortuous, but when connected form an undoubtedly tortuous curve.

Therefore, we propose a new composition property, such that a vessel $s$, combination of various segments $s_i$, will have tortuosity measure not less than any of its composing parts

$$\begin{aligned}\tau(s_i) \leq \tau(s_1 \oplus s_2 \oplus \cdots \oplus s_n)\\ \forall\, i = 1 \ldots n | s_i \subseteq s.\end{aligned} \tag{2}$$

### C. Modulation

It is useful now to express a monotonic relationship with respect to two other properties, which we will call frequency modulation at constant amplitude and amplitude modulation at constant frequency. It may be assumed that the greater is the number of changes in the curvature sign (*twist*), the more tortuous the vessel can be considered. Similarly, the greater is the amplitude (maximum distance of the curve from the underlying chord) of a segment, the greater is the tortuosity associated with it. Defining a *turn curve* as a segment between two consecutive twists, for two vessels having turn curves with the same amplitude, the difference in tortuosity varies with the number of turn curves $\varphi$

$$\varphi(s_1) \leq \varphi(s_2) \Rightarrow \tau(s_1) \leq \tau(s_2) \tag{3}$$

and, conversely, for two vessels with the same number of turn curves (with the same frequency), the difference in tortuosity depends on the difference in amplitude $\alpha$ of the turn curves

$$\alpha(s_1) \leq \alpha(s_2) \Rightarrow \tau(s_1) \leq \tau(s_2). \tag{4}$$

## III. Methods

### A. Available Tortuosity Measures

Various tortuosity measures have been proposed in the literature (see e.g., [8]), but all fail in certain aspects. In this section, we will briefly review the available methods for evaluating retinal vessel tortuosity and we will discuss their drawbacks and theoretical pitfalls in Section VI. For a more detailed description of these measures, we refer the reader to the original quoted papers.

A vessel can be described as a curve

$$s(l) = [x(l), y(l)] : D \subset \mathbb{R} \rightarrow \mathbb{R}^2 \tag{5}$$

with $s \epsilon C^1(\mathbb{R})$ and $l$ the curvilinear coordinate defined on an interval $D$ of $\mathbb{R}$.

The *chord length* $L_\chi$ is defined as the distance between the curve end points

$$L_\chi = \|s(\max(D)) - s(\min(D))\| \tag{6}$$

and the *curve length* $L_c$ is

$$L_c = \int_{\min(D)}^{\max(D)} \left\| \frac{\partial s}{\partial l} \right\| dl. \tag{7}$$

The curvature $\kappa(l)$ of $s(l)$ is

$$\kappa(l) = \frac{\frac{dx}{dl}\frac{d^2y}{dl^2} - \frac{d^2x}{dl^2}\frac{dy}{dl}}{\left(\left(\frac{dx}{dl}\right)^2 + \left(\frac{dy}{dl}\right)^2\right)^{3/2}}. \tag{8}$$

*1) Arc Length Over Chord Length Ratio:* The simpler and most widely used measure of a vessel tortuosity is the ratio between its length $L_c$ and the length of the underlying chord $L_\chi$ [5], [8], [20], [21]. It was first proposed in [21], under the assumption that the non-tortuous vessel is a straight line, and that the radius of curvature is much larger than the chord.

*2) Measures Involving Curvature:* Hart *et al.* [8] presented a number of tortuosity measures that involve the use of the integral of the absolute curvature $(tc)$ or of the squared curvature $(tsc)$. The idea behind this is that integral measures of $\kappa$ over the domain $D$ should be a measure of the variability of vessel direction

$$\begin{aligned} tc &= \int_{\min(D)}^{\max(D)} |\kappa(l)|\, dl \\ tsc &= \int_{\min(D)}^{\max(D)} |\kappa(l)|^2\, dl. \end{aligned} \tag{9}$$

Integrating along domains possibly different in dimension yields measures that depend on the aforesaid dimension. The ratios between the absolute curvature integral $tc$ (or the squared curvature integral $tsc$) and chord length $L_\chi$ (or vessel length $L_c$) have been proposed to circumvent this point.

Recently, a different curvature-based tortuosity measure has been proposed [22]. The derivative of the curvature was considered to catch the information on directional change of a line; therefore, the tortuosity is computed as the integral of the squared derivative of curvature

$$\mathrm{DCI} = \frac{1}{L_c} \int_{\min(D)}^{\max(D)} \left| \frac{d\kappa(l)}{dl} \right|^2 dl. \tag{10}$$

*3) Mean Direction Angle Change:* A measure of tortuosity based on the local direction variation of the vessel has been also proposed [23]. It computes the average of the angles between sample center points describing the vessel. Being $\boldsymbol{d_i}$ the coordinates of the $i$th center point of the vessel, and $\boldsymbol{v}_{i+n}$ the vector going from the $i$th sample to the $(i+n)$th

$$\boldsymbol{v}_{i+n} = \boldsymbol{d_{i+n}} - \boldsymbol{d_i} \tag{11}$$

$$\boldsymbol{v}_{i-n} = \boldsymbol{d_{i-n}} - \boldsymbol{d_i} \tag{12}$$

the vessel direction at each point can be obtained as

$$\theta(i) = \arccos\left(\boldsymbol{v_{i+n}} \cdot \boldsymbol{v_{i-n}}\right) \tag{13}$$

an the mean angle change metric (MAC) is then defined as:

$$MAC = \frac{1}{N - 2 \cdot n} \sum_{i=1}^{n} \theta(i) \tag{14}$$

where $N$ is the number of vessel samples, and $n$ is a fixed, user-defined parameter, which provides a regularization of the estimated direction.

*4) Absolute Direction Angle Change:* The local direction variation was used also in [24] to measure vessel tortuosity. For each center point $i$ of a vessel, the local direction change $\theta(i)$ is computed with the same method described in the previous section. The tortuosity measure TN is computed as the number of times the angle $\theta(i)$ at each center point is above a fixed threshold $th_{\mathrm{TN}}$. In [24], $th_{\mathrm{TN}} = \pi/6$, and thus

$$\mathrm{TN} = \sum_{i=1}^{n} \left( \theta(i) \geq \frac{\pi}{6} \right). \tag{15}$$

*5) Inflection Count Metric:* Bullit *et al.* [9] noted that the tortuosity measure $L_\chi / L_c$ may not distinguish between smoothly curved vessel and vessels that make abrupt changes in direction. To overcome this, they proposed to evaluate the number of *inflection points* (*twists*), as the number of points in which the Frenet normal axis to the vessel changes direction by $\pi$ radians, that is equivalent to a change in sign of the curvature for planar curves. They propose the following tortuosity measure:

$$\mathrm{ICM} = (n_{ic} + 1) \frac{L_\chi}{L_c} \tag{16}$$

where $n_{ic}$ is the number of twists.

### B. Proposed Tortuosity Measure

When evaluating tortuosity, ophthalmologists integrate information on how many times a vessel changes convexity (or curvature sign) and how large is the amplitude of each of the recognized *turn curves* (each segment delimited by two changes in convexity). We would therefore decompose any vessel into a set of consecutive segments of constant curvature sign.

Having defined the curvature of a curve as in (8), we define a segment $s_i$ of a vessel segment $s$ as a *turn curve* if

$$[\kappa(l) \geq 0,\ \forall\, l \epsilon D_i] \vee [\kappa(l) \leq 0,\ \forall\, l \epsilon D_i] \tag{17}$$

with $D_i$ a compact subset of $D$. Due to the possible presence of segments with zero curvature (straight lines), the decomposition is not unique. Moreover, these segments affect the tortuosity measure, since their length enters into the computation of turn curves amplitude. In fact, not considering these straight segments, might lead to the situation in which two turn curves linked by a straight line will yield the same tortuosity value, independently from the distance and orientation deriving from the straight connection (Fig. 3). To take this into account, we chose to split the straight segments into two halves, assigning one to the preceding and one to the following turn curve.

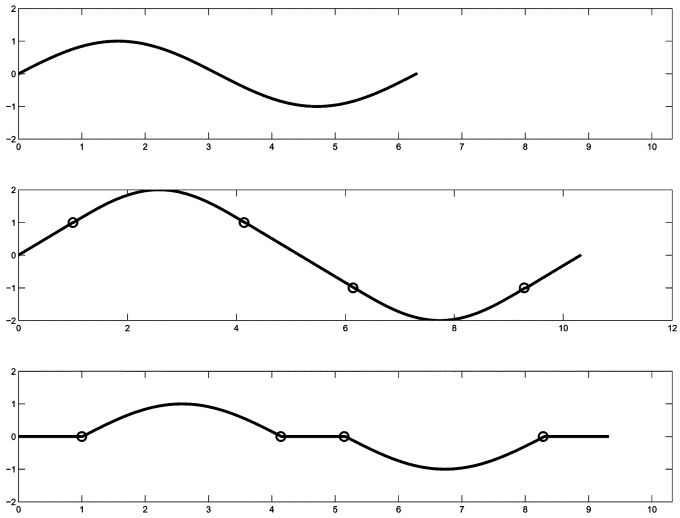


Fig. 3. Three curves with the same curvature but different aspect. In the top panel, a sinusoid is shown, and in the middle and bottom panel the same sinusoid is split in half with the two parts linked by straight lines. Circles show the limits of the two sinusoid halves.

Once a curve $s(l)$ is divided into $n$ turn curves

$$s_i : s = s_1 \oplus s_2 \oplus \cdots \oplus s_n. \tag{18}$$

we propose a new index to quantitatively describe vessel tortuosity as

$$\tau(s) = \frac{n-1}{n} \frac{1}{L_c} \sum_{i=1}^{n} \left[ \frac{L_{c_{s_i}}}{L_{\chi_{s_i}}} - 1 \right]. \tag{19}$$

It is worth noting that this measure evaluates vessel tortuosity by summing local contributions to tortuosity by assessing how much each turn curve is different from a smooth curve, at variance with all other methods (including that described in [9]), which evaluate the tortuosity globally on the entire vessel. This tortuosity measure has a dimension of $1/\mathrm{length}$ and thus may be interpreted as a *tortuosity density*, allowing its comparison on vessels of different length. Otherwise, a sinusoidal curve considered over a number of cycles would yield a lower tortuosity than that evaluated on a greater number of cycles, which is not consistent with the clinical perception of retinal vessel tortuosity. Moreover, when $n$ is equal to 1 (a vessel with no twist) then $\tau$ is equal to 0 and thus vessels with a constant convexity have zero tortuosity, whereas when more than one twist is present on a vessel, the measure is always greater than zero.

The proposed definition of tortuosity meets all the properties described in Section II. In particular, the composition property is satisfied via the summation in (19), the amplitude modulation via the ratio of the length over the chord length for every turn curve, and the frequency modulation is taken care of implicitly by means of the curve splits in (19). Even if curvature depends on the scale, its sign does not. Hence, this measure provides a unique split of a vessel into segments, regardless of its magnification and scaling. Since the length over chord ratio is scale invariant, the whole measure we propose is invariant to image magnification, as long as the total length $L_c$ is expressed

in common length units (e.g., $\mu$m), and not in image related units (e.g., pixels).

## IV. Experiment Setup

To test the performance of the proposed tortuosity index and to compare it with the other available indexes, three types of experiments have been performed. In the first one, different types of vessels have been simulated, to evaluate the compliance of the various methods with the tortuosity properties described in Section II.

In the second one, two sets of retinal vessels have been extracted from fundus images and manually traced. No automatic vessel identification algorithm was used, in order to decouple the evaluation of the tortuosity measures from the possible errors and variability arising from the different automatic vessel tracing procedures available. The vessels have been then ordered by a retina specialist by increasing tortuosity, and the correlation of the methods with the clinical perception has been computed.

Finally, a Monte Carlo simulation has been performed on the manually traced vessels, to test the degradation of correlation in presence of noisy samples.

### A. Simulated Vessels

In order to test if and how the various proposed tortuosity measures vary when single parameters that influence the clinical perception of tortuosity vary, three sets of simulated vessels have been generated.

The first set is composed by a sequence of sinusoids with the same amplitude but different frequency

$$s(l) = A \cdot \sin(f \cdot l) \tag{20}$$

with $A = 2, l \in [0, 6\pi] \subset \mathbb{R}$, $f \in \{0, \pi/12, 2\pi/12, \ldots, \pi/2\}$.

The second set of simulated vessels is composed by a sequence of sinusoids with the same frequency and chord length but different amplitude: $A \in [0, 1, 2, 3, 4, 5]$, $l \in [0, 6\pi] \subset \mathbb{R}$, $f = \pi$.

The third and last set of simulated vessels is composed of curves with constant curvature (in absolute value), but increasing number of changes of curvature sign. These curves are built as follows. A semi-circumference of radius $r = 4$ is incrementally divided into a number of arcs (from 1 to 6), which are then arranged in a curve: adjacent arcs are linked in such a way to have different curvature sign and the same first and second derivative at the joint: the obtained set of curves is shown in Fig. 4.

### B. Manually Traced Vessels

A set of 60 retinal images, from different eyes of 34 subjects with normal retina or affected by hypertensive retinopathy of different severity, were acquired with a fundus camera with a 50° field-of-view and then digitized with a scanner at 1300×1100 pixels. From these images, a set of 30 arteries and a set of 30 veins of similar length and caliber have been extracted. They were chosen as the major arteries or veins that have minimal overlap or entangling with surrounding vessels, so as to provide a minimally confusing view. The center line

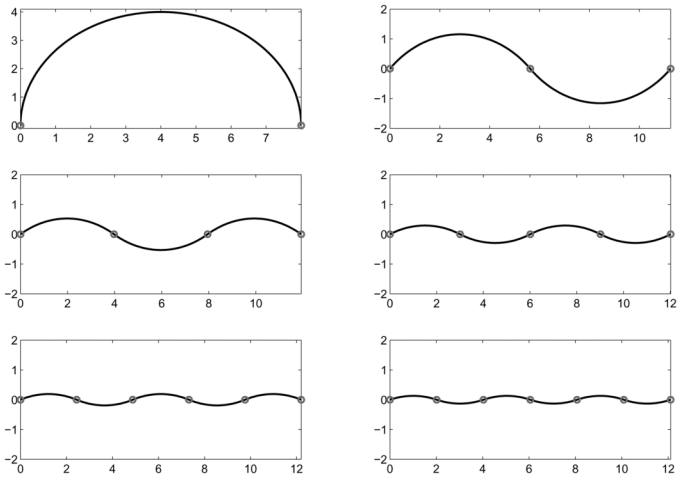


Fig. 4. Simulated vessels obtained by breaking a semi-circular arc into smaller arcs, and arranging them so that each arc has curvature sign different from the preceding one. The circles represent the points of changing curvature sign.

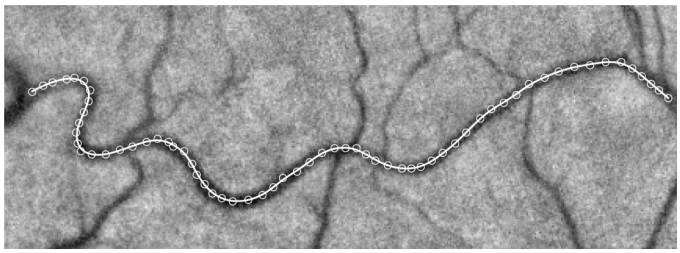


Fig. 5. Vessel in the arterial set with the manually placed sample center points (white dots) and the smoothed center line $s(l)$.

of the selected vessels have subsequently been traced, by manually placing a number of sample center points along the vessel course.

Being the sampling procedure user-dependent, it may lead to a sparse vessel description (few samples), and thus a poor description of vessel dynamics, which may eventually fail to provide the necessary information on vessel direction and its derivatives. Sparseness of data was taken care of by means of an interpolation function, to describe the vessel between sampling points, where no data are present. The fact that vessels are continuous structures at least with their first derivative led to use a cubic smoothing spline interpolation.

Data have thus been interpolated through the smoothing spline (Fig. 5), with the regularization parameter set to $\gamma = 0.005$, to obtain two sets of 30 interpolated vessels (one artery set and one vein set), on which all the tortuosity measures previously described were computed. At variance with the case of simulated curves, when dealing with real vessels it is common to find small oscillations (changes of convexity with very small amplitude) around the main vessel direction, due to the presence of noisy samples. These oscillations might affect the correct decomposition of the vessel, creating a great number of artificial turn curves, which are not clinically relevant to the retinopathies considered. An hysteretic threshold on curvature was thus used to deal with these small variations, as shown in Fig. 6: the vessels are not split where their curvature crosses the zero line, but in the midpoint of two hysteresis-crossing points.

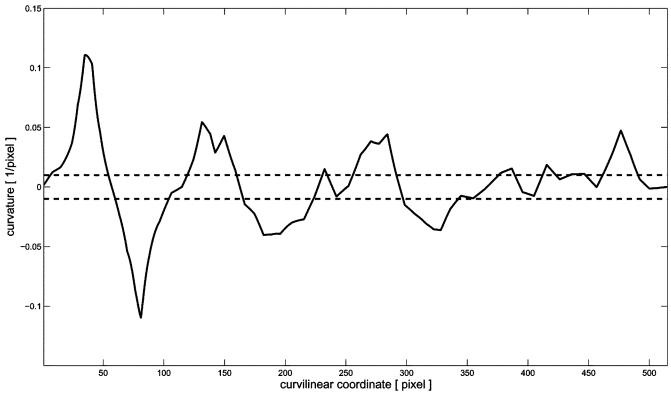



Fig. 6. Curvature value along a vessel. Dashed lines indicate the hysteretic threshold for the convexity change, in this case of an artery (see Section IV-B).

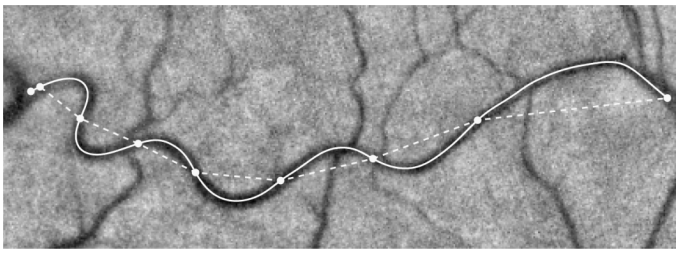


Fig. 7. Vessel in the arterial set with superimposed the filtered center line $s(l)$. Dots represent the points of vessel splitting into segments of equal curvature sign, and the dashed lines represent the chord corresponding to each turn curve.

An example of the application of this procedure can be seen in Fig. 7, where the identified twists are linked by a straight line.

A retina specialist ordered by increasing tortuosity the vessels in the two sets separately: in this way, not only the capability of the proposed methods to capture the pathological course can be assessed, but also their capability to capture its severity.

### C. Robustness Evaluation

The experiments of the previous section were performed in the ideal setting of "exactly" traced vessels. This is not the case when the measures has to be provided by an automatic diagnostic tool, which is the aim of many developed softwares for ophthalmology. Automatic vessel tracing procedures are subject to image noise, shadows, vessels overlapping, and other disturbances. Even the skeletonization of automatically segmented vessel pixels is prone to provide with noisy center lines. This situation may be modelled by describing the manual vessel samples $s(k)$ as a stochastic variable with mean corresponding to vessel center line $s(l_k)$, and variance depending on an additive noise component

$$\boldsymbol{s}(k) = \boldsymbol{s}(l_k) + \boldsymbol{e}(k) \tag{21}$$

$$\boldsymbol{e}(k) \sim \mathcal{N}(0, \Sigma) \tag{22}$$

with $\boldsymbol{e}(k)$ the two dimensional vector representing the noisy displacement along the $x$ and $y$ axis of the sample from the manual center point. The variance matrix $\Sigma$ used in the setup was diagonal with $\Sigma_{11} = \Sigma_{22} = 0.5$ and null off-diagonal terms. Under this assumption, the noise on the two coordinates

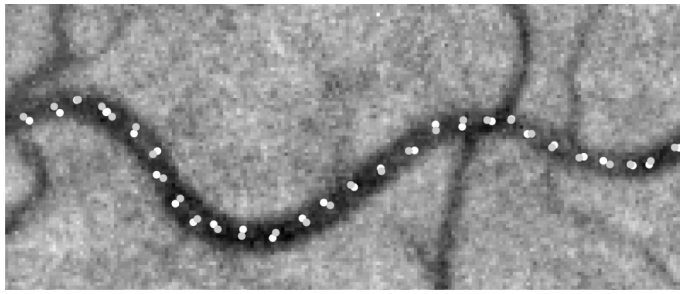


Fig. 8. Two different center line samples for the same vessel, obtained with the procedure described in Section IV-C. Gray dots represent the manually placed samples, whereas the white dots are the manual samples with added noise.

is independent, and a noisy center point $s(k)$ lays with probability 0.99 within a circle of radius 2.12 pixels, centered on $s(l_k)$ (see Fig. 8). This is a reasonable model for the large retinal vessels we are considering, under the assumption that an automatic vessel recognition algorithm will not badly misplace center points out of the true course, but that center points can be affected by small fluctuation within the vessel borders. With this model, a Monte Carlo simulation can be run to obtain noisy data that simulate those provided by an automatic algorithm, allowing the evaluation of the robustness to noise of the various tortuosity measures.

Starting from the two sets of manually traced vessels, 100 noisy repetitions of these sets have been generated. The smoothed version of the vessels is then obtained from this noisy samples, and the correlation of tortuosity measures with the clinician grading was calculated.

## V. Results

The different tortuosity measures evaluated on the simulated vessel sets described in Section IV-A are shown in Figs. 9–11.

In Fig. 9, the tortuosity measures as functions of the frequency are shown: since the sinusoids have the same chord length and the same peak elevation, tortuosity measures should increase with increasing frequency.

In Fig. 10, the various tortuosity measures as functions of the sinusoid amplitude are shown. Since only the sinusoid amplitude changes, a correct tortuosity measure is expected to increase with increasing amplitude.

In the third set, vessels with increasing number of curvature-sign changes are simulated, while the absolute value of the curvature and the total curve length remain constant, whereas chord length, peak elevation, and frequency vary. The different tortuosity measures as function of the number of twists are shown in Fig. 11. Even if it is not clear how a tortuosity measure should account for the aforementioned vessel variations, the tortuosity perception is however different for the various curves, and, in the considered retinopathies framework, the lowest tortuosity value should be measured on the semicircular curve (top-left panel of Fig. 4).

In this simulated vessels analysis, only six of the twelve measures ($\tau$, $tc$, $tsc$, $tsc/L_c$, ICM, and DCI) increase in all cases when the amplitude, the frequency or the number of twists increase, and are hence the best candidates to correctly match the ophthalmologist's grading.

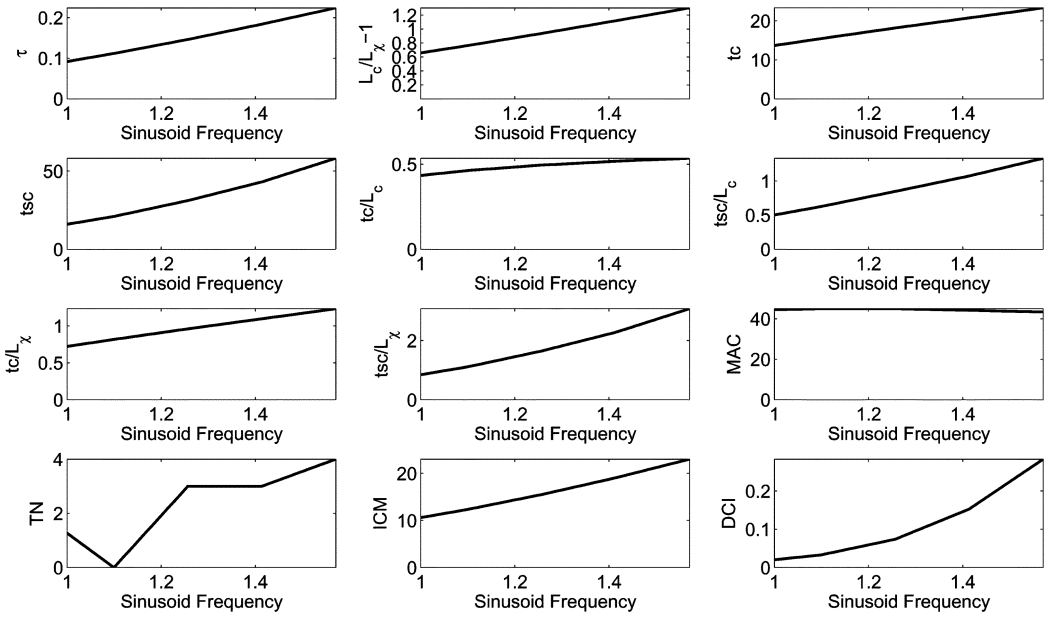



Fig. 9. Tortuosity indexes as a function of the number of twists for the first simulated vessel set.

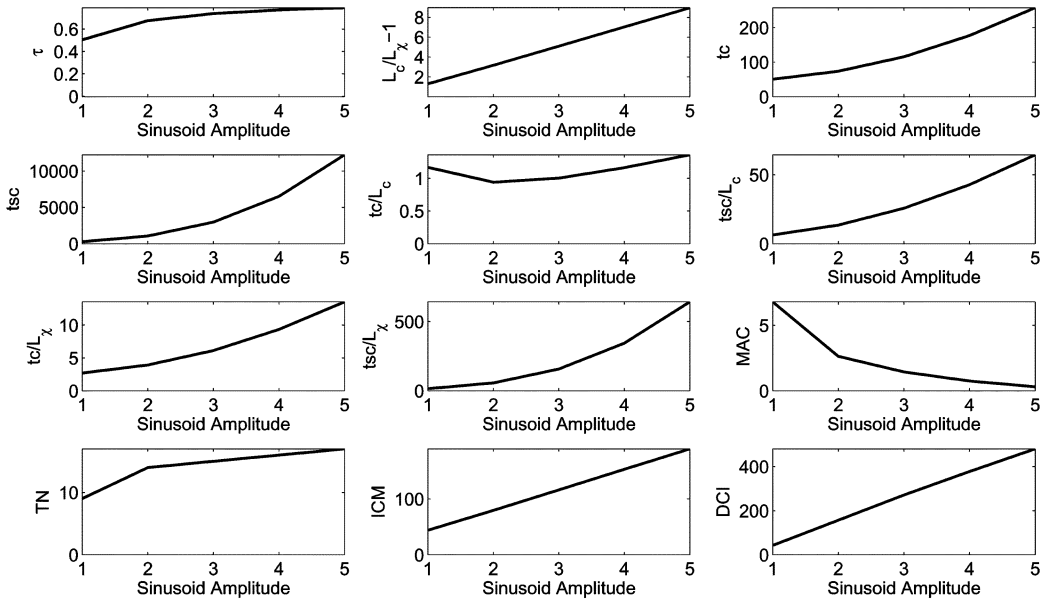



Fig. 10. Tortuosity indexes as a function of the amplitude, for the second simulated vessels set.

For the two sets of manually traced vessels, the values of the correlation with the clinician's grading for all the tortuosity indexes considered in this paper are shown in Table I. Since we are considering correlations among rankings, Spearman correlation $\rho$ is used. All the obtained correlations were significant (under the null hypothesis $H_0 : \rho = 0$, $p$-value is $< 0.0001$ at a significance level $\alpha = 0.05$). These results show that the proposed measure $\tau$ has the best correlation, as regard both artery and vein tortuosity.

These results are confirmed by the analysis of the robustness evaluation described in Section IV-C, whose results are reported in Table II. The histograms of the correlations over the 100 repetitions can be seen in Fig. 12 for arteries and Fig. 13 for veins. They show that the proposed measure $\tau$ has a median value always larger than the median of all other algorithms, and a variance smaller than all but the $L_c/L_\chi$ ratio, as summarized in Table II.

## VI. Discussion and Conclusion

Even if the tortuosity measure $L_c/L_\chi$ has been widely used, it contains the implicit assumptions that the nontortuous vessel is a straight line and that the radius of curvature is much larger than the chord. The first assumption holds when the vessels are not constrained to lie on a nonplanar shape, which happens in the retina, whereas the second is not met in very tortuous vessels.

The intuitive meaning of this ratio is that the greater its value, the more distant the vessel is from a straight line, i.e., tortuous: being the surface of the retina close to a hemisphere, the non

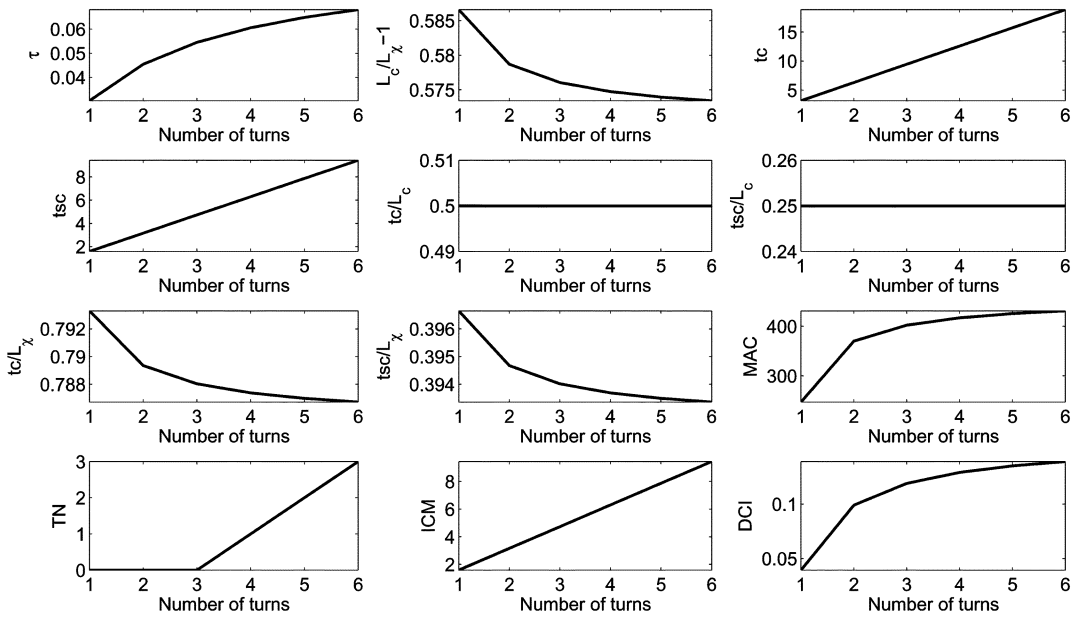



Fig. 11. Tortuosity indexes as a function of the number of the composing circular arcs, for the third simulated vessels set.

TABLE I
SPEARMAN CORRELATION COEFFICIENTS OF THE TORTUOSITY MEASURES WITH THE OPHTHALMOLOGIST GRADING

| Tortuosity Measure | Arteries | Veins |
|---|---|---|
| $L_c/L_\chi$ | 0.792 | 0.656 |
| tc | 0.922 | 0.837 |
| tsc | 0.925 | 0.826 |
| $tc/L_c$ | 0.919 | 0.814 |
| $tsc/L_c$ | 0.917 | 0.773 |
| $tc/L_\chi$ | 0.939 | 0.842 |
| $tsc/L_\chi$ | 0.928 | 0.804 |
| $MAC$ | 0.920 | 0.814 |
| $TN$ | 0.838 | 0.695 |
| $ICM$ | 0.684 | 0.575 |
| $DCI$ | 0.787 | 0.589 |
| $\tau$ | 0.949 | 0.853 |

TABLE II
MEDIAN (STANDARD DEVIATION) OF THE SPEARMAN CORRELATION OF THE TORTUOSITY MEASURES WITH THE OPHTHALMOLOGIST GRADING OVER 100 SIMULATED NOISY VESSELS

| Tortuosity Measure | Arteries | Veins |
|---|---|---|
| $L_c/L_\chi$ | 0.793 (0.006) | 0.659 (0.012) |
| tc | 0.909 (0.012) | 0.752 (0.380) |
| tsc | 0.918 (0.008) | 0.752 (0.047) |
| $tc/L_c$ | 0.918 (0.009) | 0.705 (0.060) |
| $tsc/L_c$ | 0.913 (0.007) | 0.661 (0.060) |
| $tc/L_\chi$ | 0.935 (0.008) | 0.779 (0.044) |
| $tsc/L_\chi$ | 0.920 (0.007) | 0.710 (0.056) |
| $MAC$ | 0.915 (0.010) | 0.703 (0.060) |
| $TN$ | 0.826 (0.018) | 0.688 (0.073) |
| $ICM$ | 0.582 (0.083) | 0.259 (0.108) |
| $DCI$ | 0.792 (0.029) | 0.500 (0.067) |
| $\tau$ | 0.945 (0.006) | 0.804 (0.044) |

tortuous paradigm should be the circle arc (including the degenerate case of circle arc with infinite radius of curvature). In fact, vessels with a constant and small curvature, regardless of the amplitude of the arc it describes (as for the main retinal vessels), will be regarded by an ophthalmologist as of negligible tortuosity. The same considerations hold if other models for the vessel arcades are assumed, as the arc of ellipse or the parabola as in [25]. Moreover, it is shown in Fig. 2(a) and (b) that two vessels with very different tortuosity can have the same $L_c/L_\chi$ measure.

Even if the proposed method can not distinguish tortuosities of vessels with constant curvature, but possibly different amplitude (shallow C-shaped vessels with respect to deep U-shaped vessels), these are conditions rarely seen in retinal vessels.

Tortuosity measures based on integrals involving the curvature without taking into account its sign pose another problem. The examples presented in Fig. 2(b) and (c) show that a smaller curvature integral may correspond to a greater perceived tortuosity. In our opinion, there are two main reasons for this result. The first is that curvature is different from zero only along arcs, whereas it is negligible along straight or almost straight segments; but straight segments together with arcs dramatically change the tortuosity appearance. The second is that changes in convexity of the curve (curvature sign) are not taken into account. On the contrary, this is the main feature used by expert graders to assess tortuosity.

The same drawbacks affect tortuosity measures involving the computation of local angle change (MAC and TN). Even the simple examples of Fig. 2 show that a vessel with constant curvature, such as a semi-circumference [Fig. 2(b)] and a vessel formed by the juxtaposition of two arcs of circumference [Fig. 2(c)] have the same average angle variation, despite their difference in tortuosity. Moreover, also for this type of measures, vessel segments with no variation in direction do not contribute to the tortuosity measure. The situation is further

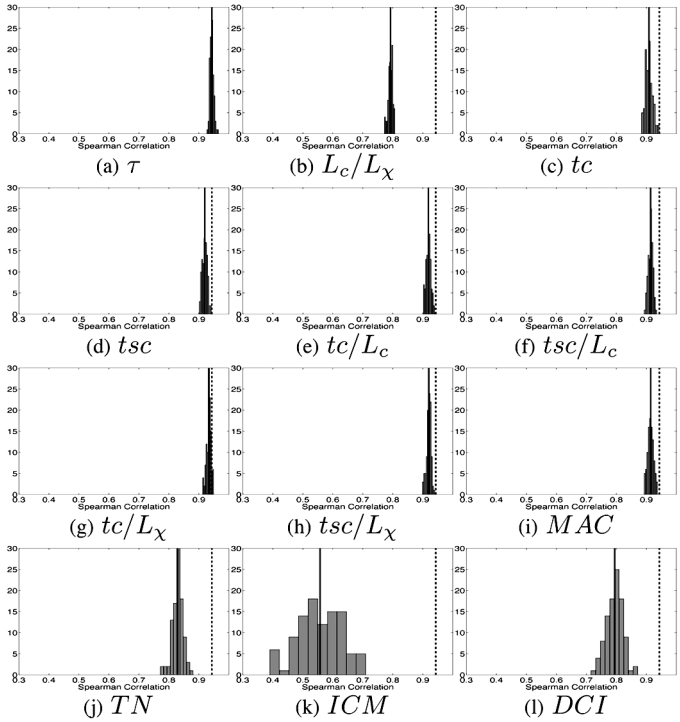



Fig. 12. Histograms of the Spearman correlation values for the various tortuosity measurements on the set of arteries obtained by corrupting with noise the manually traced vessel center points. In each panel, the solid vertical line is the median value of the histogram, and the dashed vertical line is the median of the values of the proposed tortuosity $\tau$, shown for comparison.

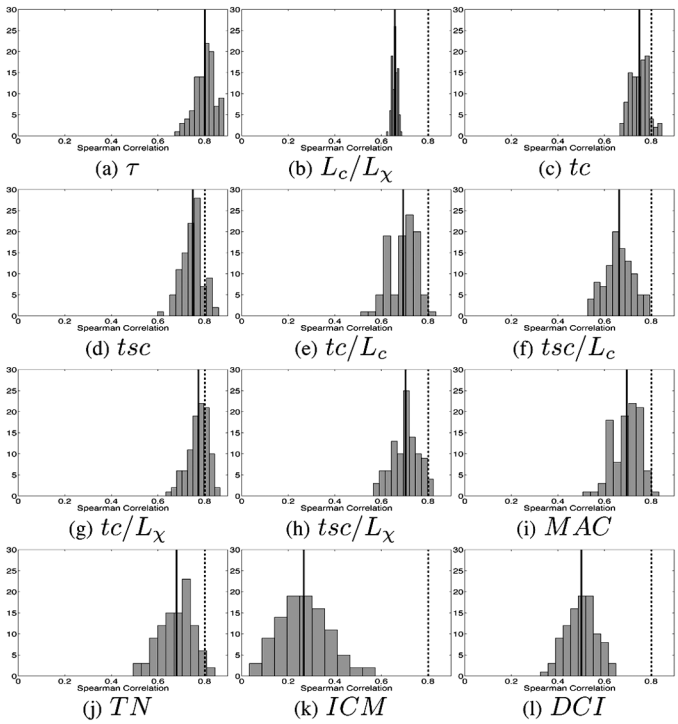



Fig. 13. Histograms of the Spearman correlation values for the various tortuosity measurements on the set of veins obtained by corrupting with noise the manually traced vessel center points. In each panel, the solid vertical line is the median value of the histogram, and the dashed vertical line is the median of the values of the proposed tortuosity $\tau$, shown for comparison.

exacerbated by the hard threshold on the local direction in case of TN. The theoretical drawbacks of these methods are confirmed by the results obtained in the experiments, especially in those regarding robustness evaluation.

Even the measure taking into account the number of twists, ICM, fails in reflecting the clinician's perception of tortuosity. This may suggest that the human evaluation of tortuosity of retinal vessels is based more on the local winding of a vessel than on a global difference between a straight line and the curve under examination.

The measure we propose in this work is the only one among those considered that provides this local evaluation, which appears to be able to distinguish between tortuous and smoothly curved vessels. A limitation of the proposed measure is its scale-dependency. Since it is a density measure, two vessels with the same geometry, but differing only in scaling, will result in tortuosity measures scaled correspondingly. Even if this is not a problem when comparing vessels with similar length and caliber, it might become a problem when a global measure of tortuosity for the entire retinal image has to be computed. In this case, short vessels with high tortuosity may boost up the whole retinal image tortuosity, despite their little impact on the length of the whole retinal vessel structure. This is however a complex problem *per se*, which is beyond the scope of this paper. It needs careful clinical studies, in which an appropriate weighting scheme, e.g., based on the vessel caliber, should be investigated to obtain a clinically sound combination of single vessel tortuosities into an image tortuosity.

In retinal images, vessels cross, bifurcate, and branch, so that each vessel analysis algorithms need to correctly identify these different situations, since they lead to different connectivity of the involved segments. Failing in connecting two segments, or conversely linking the wrong segments, may lead to inconsistent tortuosity measure. Even if this topic is beyond the scope of this paper, it is worth noting that for an automatic vessel tortuosity measure, it is important that crossing, bifurcations and branches are correctly identified, as proposed in e.g., [18], [19], so that vessel course may be fully reconstructed.

In this paper, we have proposed a new algorithm for evaluating tortuosity of retinal vessels able to solve the theoretical drawbacks of some of the algorithms proposed in the literature, arising from the particular geometry of the retina. All the methods that, to the best of our knowledge, were proposed in the literature to evaluate retinal vessel tortuosity have been implemented and comparatively evaluated in the present work. The algorithm we propose proved to be the most accurate to describe the ophthalmologist's perception of retinal vessel tortuosity, in the case of hypertensive retinopathy. The proposed method, along with the $L_c/L_\chi$ measure, also proved to be the most resilient to a sparse and noisy description of a vessel, which makes it well suited for implementation as a tortuosity assessment module in an automatic software for retinal vessel analysis.

Even if we tested the tortuosity measures with vessel changes due to hypertension, investigations are being conducted to assess the goodness of the proposed measure in other retinopathies

that show similar vascular morphology changes, such as diabetic retinopathy and retinopathy of prematurity.

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